

SemLiFi: Burst-Aware Semantic Recovery With Confidence-Gated Fallback for LiFi Links

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Abstract—Visible light communication (VLC) and Light Fidelity (LiFi) links are highly sensitive to temporary line-of-sight blockage. Unlike independent bit errors, a physical obstruction can remove a contiguous interval of symbols, producing a burst loss whose duration and location depend on obstruction dynamics and link geometry. This paper presents SemLiFi, a cross-layer burst-recovery framework that combines a physically implemented LiFi testbed, controlled optical occlusion, empirical burst characterization, Burst-Aware Semantic Reconstruction (BASR), and a Confidence-Gated Fallback Protocol (CGFP). BASR reconstructs missing structured telemetry from surviving context and burst information, while CGFP routes each reconstruction either to semantic acceptance or to selective exact recovery. The prototype uses ESP32 transmit and receive nodes, on-off keying with Manchester coding, a high-intensity LED, a BPW34 photodiode, an LM358 transimpedance amplifier, CRC-8 validation, and an SG90 servo-controlled optical flap. The reported study contains 56 empirical occlusion events, a 74,281-parameter edge-constrained Transformer BASR model, and disjoint 664/641-burst calibration/test splits for CGFP. The clean-link path reports 449.0 ms mean latency with zero bit errors. The evaluated Reed–Solomon configuration recovers 37.5% of observed occlusion bursts, while Stop-and-Wait ARQ reports 97.1% delivery at 1560.0 ms mean latency. BASR reports $42.63 \pm 1.47\%$ exact reconstruction accuracy and 100.0% accuracy on the evaluated motor-state field. At the calibrated threshold $\tau^* = 0.80$, accepted semantic patches have 97.13% precision and a 0.78% patch-error rate, with 27.15% of retransmissions avoided in the reported simulation. Continuous-burst simulation reaches 100% delivery at 790.7 ms with $1.00 \pm 0.13\%$ UFER, while a 15-frame live run reaches 100% delivery and 0.00% UFER by rejecting all low-confidence semantic reconstructions and invoking exact fallback. The results support the feasibility of the integrated recovery architecture within the tested structured-telemetry and controlled-occlusion setting, while the small live-hardware sample and absence of completed component ablations limit broader claims.

Index Terms—LiFi, visible light communication, optical wireless communication, burst errors, semantic communication, semantic reconstruction, confidence gating, edge inference, embedded Transformer, error-resilient telemetry.

I. INTRODUCTION

Visible light communication (VLC) uses controlled modulation of optical intensity to convey information, while LiFi extends optical wireless communication toward bidirectional and networked connectivity. VLC has been investigated for indoor communications because of the availability of optical

spectrum, electromagnetic compatibility, spatial confinement, and the possibility of reusing lighting infrastructure. Foundational channel analyses and subsequent surveys established the importance of transmitter–receiver geometry, irradiance angle, receiver orientation, optical filtering, and line-of-sight conditions in determining link quality [1]–[3]. IEEE 802.15.7 further provides a standards framework for short-range optical wireless communication [4].

The same line-of-sight dependence that is useful for optical spatial confinement creates a distinctive reliability problem. A person crossing an optical path, a temporary object, equipment movement, receiver rotation, or another obstruction can attenuate the received signal for a finite interval. Consequently, errors need not be independent in time. Instead, several adjacent symbols or bytes can become unavailable together. For a physical system, the resulting burst length is linked to obstruction duration, optical geometry, modulation timing, and receiver thresholding. A recovery mechanism that assumes only independent random errors can therefore fail to represent the dominant impairment mode of an obstructed optical link.

Conventional error-control mechanisms remain essential in this setting. Forward error correction (FEC) adds redundancy so that a receiver can recover corrupted information without a retransmission. Reed–Solomon (RS) codes are a classical family of block codes with well-established symbol-error and burst-error decoding properties [5], [6]. Their correction capability, however, is bounded by the redundancy selected for the code. Automatic repeat request (ARQ) instead requests missing or corrupted information again and can preserve exact payload content, but the retransmission itself consumes time and requires a return path. When the forward optical path is temporarily unavailable, retransmission latency can become significant.

Semantic communication offers a complementary perspective. Rather than treating every source symbol as equally important, semantic systems exploit source structure, context, or task objectives to infer information that is useful to the receiver [7]–[9]. Recent work has also demonstrated semantic communication over visible-light links and underwater optical wireless channels, showing that learning-based semantic methods can be coupled to physical optical communication systems [10], [11]. These developments motivate a narrower question: can contextual semantic reconstruction be used specifically for temporary optical burst loss while preserving an exact-recovery path whenever the reconstruction is not sufficiently trustworthy?

This paper presents SemLiFi, a cross-layer architecture designed around that question. SemLiFi does not claim that semantic reconstruction should replace FEC or ARQ. Instead,

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it introduces a conditional recovery path. First, the physical link generates and measures a burst caused by controlled optical occlusion. Second, Burst-Aware Semantic Reconstruction (BASR) estimates the missing structured payload from surviving context and burst metadata. Third, a Confidence-Gated Fallback Protocol (CGFP) combines semantic confidence, physical burst severity, and structural validity. A reconstruction is accepted only when it satisfies the calibrated decision criterion; otherwise, the receiver requests selective exact recovery.

The research question is therefore: *Can a physically implemented LiFi link exploit contextual semantic reconstruction during temporary burst loss while retaining an exact-recovery path for uncertain predictions?*

The contributions are fourfold:

- 1) a low-cost ESP32-based LiFi testbed with repeatable servo-controlled optical occlusion;
- 2) a burst-aware reconstruction methodology in which corruption patterns are derived from measured optical burst behavior rather than only independent random masking;
- 3) a confidence-gated routing mechanism combining model confidence, physical burst severity, and hard structural validity; and
- 4) an evaluation framework comparing semantic recovery and exact fallback with no recovery, Reed–Solomon FEC, and Stop-and-Wait ARQ.

The remainder of the paper is organized as follows. Section II reviews related work and identifies the research gap. Section III defines the notation and system model. Section IV describes the physical testbed and burst characterization. Section V presents BASR and CGFP. Section VI describes the experimental methodology. Section VII reports the measured and simulated results available in the supplied experimental record. Section VIII discusses the findings, limitations, and threats to validity. Section IX gives reproducibility details and future work, followed by the conclusion.

II. RELATED WORK AND RESEARCH GAP

A. Optical Wireless Communication

The physical behavior of VLC links has been extensively studied. Komine and Nakagawa developed a foundational analysis of indoor VLC links, including optical channel gain and geometric factors [1]. Elgala *et al.* surveyed indoor optical wireless communication and highlighted practical issues associated with optical front ends, modulation, and link design [2]. Pathak *et al.* subsequently reviewed VLC communication, networking, and sensing, identifying challenges spanning the physical, networking, and application layers [3]. These studies establish the physical context in which temporary line-of-sight blockage can be treated as a channel impairment.

The present work focuses on a narrower failure mode: a temporary physical obstruction that creates a localized, contiguous loss interval. The objective is not to develop a new optical propagation model. Instead, the measured burst behavior is used as an input to the recovery layer. This distinction is important because the burst distribution becomes part of the

semantic recovery problem rather than merely a post hoc stress test.

B. FEC and ARQ

Reed–Solomon coding was introduced as a finite-field polynomial code and has subsequently been used for reliable communication systems requiring symbol-error correction [5]. Burst-error decoding methods further demonstrate how code structure can be exploited when errors occur in clusters [6]. In SemLiFi, RS-FEC is retained as an exact-recovery baseline. The evaluation uses RS(33,25), for which the selected configuration has $t = 4$ symbol-error correction capability.

ARQ provides a different exact-recovery mechanism. A receiver detects an invalid frame and requests retransmission. The approach is conceptually simple and remains appropriate when exact source reproduction is mandatory. The drawback considered in this work is recovery delay during a temporary optical outage. SemLiFi therefore treats ARQ as the fallback path rather than as a mechanism to be eliminated.

C. Semantic Communication

Deep-learning semantic communication has explored the use of learned representations to communicate information relevant to a receiver’s task rather than enforcing bit-perfect reconstruction in every setting [7]. Qin *et al.* discuss semantic communication principles and challenges, while Gündüz *et al.* survey context-, semantic-, and task-oriented communication architectures [8], [9]. These works motivate the use of contextual information but do not specifically address the combination of a physically measured optical burst, a localized structured-payload reconstruction task, and an exact fallback gate.

Ma *et al.* recently demonstrated a visible-light semantic communication system with a physical VLC platform and investigated semantic throughput under practical VLC constraints [10]. Nennouche *et al.* investigated semantic underwater optical wireless communication using a variational autoencoder and evaluated robustness against underwater optical channel impairments [11]. SemLiFi addresses a different recovery objective: it focuses on temporary localized burst loss and on deciding when a reconstructed telemetry segment should be trusted.

D. Research Gap

The research gap considered here is the intersection of three requirements:

- 1) the impairment is generated by a physical optical obstruction;
- 2) semantic reconstruction is explicitly conditioned on the resulting burst characteristics; and
- 3) semantic output is not trusted unconditionally but is passed through a calibrated confidence gate with an exact-recovery fallback.

The contribution is therefore architectural and experimental rather than a claim that VLC, semantic communication, or FEC is individually new. The evaluation also separates simulated

TABLE I
POSITIONING OF SEMLiFi RELATIVE TO SELECTED RELATED WORK

System	Physical optical HW	Burst-conditioned reconstruction	Exact fallback gate
Ma <i>et al.</i> [10]	Yes	No	No
Nennouche <i>et al.</i> [11]	Underwater optical	No	No
Xie <i>et al.</i> [7]	No	No	No
Qin <i>et al.</i> [8]	Survey	No	No
SemLiFi	ESP32/OOK	Yes	Yes

semantic efficiency from live hardware safety behavior, which is necessary because the available live experiment contains only 15 frames and all semantic reconstructions were rejected.

III. SYSTEM MODEL AND ARCHITECTURE

A. Notation

Let the transmitted structured payload be

$$Y = [y_1, y_2, \dots, y_N]. \quad (1)$$

An optical obstruction produces a binary corruption mask

$$B = [b_1, b_2, \dots, b_N], \quad (2)$$

where $b_i = 1$ denotes a corrupted or unavailable symbol. The physical corruption process is represented as

$$Y' = C(Y, B). \quad (3)$$

When the receiver detects a contiguous loss interval, the surviving prefix and suffix are denoted by Y^- and Y^+ . BASR estimates the missing segment as

$$\hat{Y}_m = f(Y^-, Y^+, B; \theta), \quad (4)$$

where θ denotes the learned parameters. A probabilistic formulation is

$$\hat{y} = \arg \max_y P(y \mid Y^-, Y^+, B). \quad (5)$$

The explicit use of B differentiates SemLiFi from a model that treats all missing regions as statistically identical.

The physical burst duration is

$$T_B = t_e - t_s, \quad (6)$$

and the approximate number of affected bits is

$$N_B \approx \frac{T_B}{T_b}, \quad (7)$$

where T_b is the measured bit period. The normalized burst span can be written as

$$\rho_B = \frac{N_B}{N_{\text{payload}}}. \quad (8)$$

B. Cross-Layer Recovery Pipeline

The complete pipeline is

$$\text{Physical Burst} \rightarrow \text{BASR} \rightarrow \text{CGFP}. \quad (9)$$

The physical layer first detects and characterizes the loss. BASR then estimates the missing structured content. CGFP decides whether to accept the reconstruction or request exact recovery. This ordering prevents the semantic module from operating without knowledge of the severity of the physical impairment.

TABLE II
EXPERIMENTAL CONFIGURATION USED IN THE REPORTED STUDY

Parameter	Reported configuration
TX/RX MCU	ESP32 / ESP32
Optical source	High-intensity LED, GPIO 23, OOK
Photodetector	BPW34 PIN photodiode, GPIO 34
ADC	LM358-based TIA
Amplifier	Manchester coding
Line coding	1000 μ s / 1000 bps
Bit period / rate	CRC-8, polynomial 0x31
Integrity check	SG90 servo flap, 3 profiles
Occlusion	Structured telemetry
Payload	56
Empirical burst events	74,281
BASR parameters	664 / 641 bursts
CGFP calibration/test	42, 101, 2024, 7, 99
Simulation seeds	

C. Frame and Recovery Semantics

The implemented frame uses the structure

$$[0xAA \ 0x55][LEN][SEQ][PAYLOAD][CRC8][\r\n].$$

The synchronization word is two bytes. The length field identifies the payload length and the sequence identifier supports selective retransmission. The payload is a structured telemetry string. CRC-8 uses the Dallas/Maxim polynomial $x^8 + x^5 + x^4 + 1$ (0x31). CRC validation is used for exact bit-level integrity; it is not treated as evidence that a semantic reconstruction is correct.

IV. PHYSICAL TESTBED AND BURST CHARACTERIZATION

A. Hardware Configuration

The transmitter contains an ESP32 microcontroller, a high-intensity LED, and a transistor-based LED driver. The source implementation identifies GPIO 23 as the LED control output and uses on-off keying (OOK) with Manchester coding. The receiver uses a BPW34 silicon PIN photodiode connected to an LM358-based transimpedance amplifier (TIA), followed by an ESP32 ADC input identified as GPIO 34. A software OOK slicer and Manchester decoder recover the frame. An SG90 servo-controlled flap is positioned between the optical transmitter and receiver to create repeatable obstruction events.

The component choices are intended to make the testbed low-cost and inspectable. The ESP32 provides the GPIO, ADC, timing, and processing resources needed by the prototype. The BPW34 provides a simple optical detector, while the LM358 TIA converts the photodiode current into a measurable voltage. The servo flap allows the duration and repetition of physical blockage to be controlled more consistently than an uncontrolled hand obstruction.

B. Optical Modulation and Sampling

The reported implementation uses a 1000 μ s bit period, corresponding to a raw rate of 1000 bps. Receiver sampling is performed within a mid-bit window of approximately 450–550 μ s into the bit interval. The mid-bit strategy is intended to reduce sensitivity to transition transients and timing uncertainty.

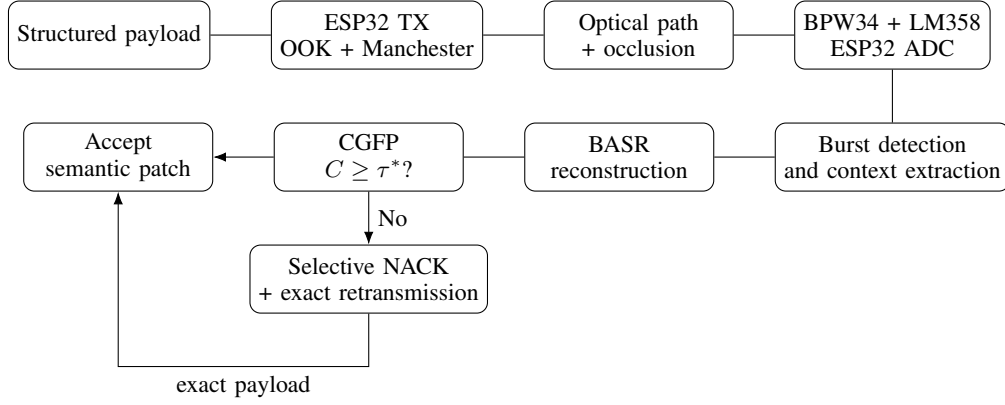


Fig. 1. SemLiFi cross-layer architecture. Physical burst observations condition BASR, and CGFP routes the reconstruction either to semantic acceptance or to selective exact recovery.

The use of Manchester coding provides a transition structure that can assist clock recovery and invalid-transition detection.

The physical system should not be interpreted as a commercial LiFi throughput benchmark. The rate is a proof-of-concept setting chosen for transparent physical recovery experiments. The paper therefore evaluates recovery behavior at the implemented rate rather than claiming performance at the much higher rates reported by specialized LiFi systems.

C. Receiver Calibration

The reported receiver calibration measured dark ADC values of approximately 0–5 counts and illuminated values of approximately 95–176 counts. An operating threshold of approximately 50 ADC counts was used. Manchester transition consistency checks complemented CRC validation. These measurements establish a substantial separation between the observed dark and illuminated levels under the reported test conditions, but the calibration is not assumed to remain constant under arbitrary ambient illumination or geometry.

D. Burst Profiles

A burst is represented as a contiguous interval in which the optical path becomes unavailable or unreliable. The logger records start time, end time, duration, corrupted symbols, affected bytes, frame identifier, and local signal information. The study reports 56 empirical occlusion events across three servo-driven profiles: P1 short blockage of approximately 15–40 ms, P2 long blockage of approximately 150–380 ms, and P3 mixed/randomized duration.

The empirical burst distribution is important for BASR. Instead of masking independent characters with arbitrary probabilities, the model is exposed to contiguous losses whose duration is related to the physical obstruction process. The source reports that physical obstructions corrupted approximately 10–25 contiguous bytes, which is substantially larger than the four-symbol correction capability of the selected RS(33,25) configuration in the evaluated cases.

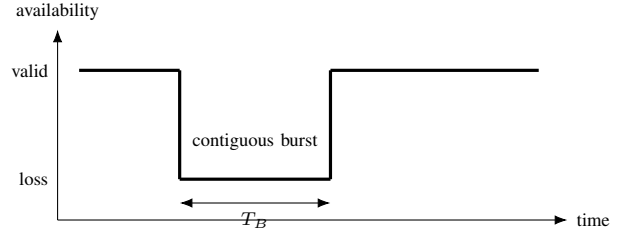


Fig. 2. Burst model used by SemLiFi. A temporary obstruction removes a contiguous interval of the optical signal; the measured duration T_B is mapped to an estimated affected-symbol count.

E. Physical Failure Modes and Threat Model

The tested impairment is non-adversarial, physically generated burst loss. The system does not claim protection against intentional frame forgery, telemetry spoofing, or an attacker who injects syntactically valid false data. Three physical cases remain particularly important: bursts longer than the training distribution, repeated or overlapping obstructions outside the measured profiles, and timing ambiguity near a CRC boundary. These cases are relevant because the confidence model is calibrated to the observed channel rather than to an arbitrary worst-case distribution.

V. BURST-AWARE SEMANTIC RECONSTRUCTION

A. Structured Payload Assumption

BASR is intended for telemetry in which fields have syntax and contextual regularities. Examples used in the study include

- TEMP=27.4, HUM=61, MOTOR=ON,
- DEVICE=04, TEMP=26.8, STATUS=ACTIVE, and
- VOLT=3.72, CURR=0.42, STATE=OK.

The structured format is central to the method. A receiver can exploit field names, delimiters, neighboring values, and application constraints to generate a candidate reconstruction. This does not imply that arbitrary encrypted binary payloads can be reconstructed. The method is evaluated only on the structured telemetry setting described by the source.

B. Edge-Constrained Transformer

BASR is implemented as an edge-constrained Transformer encoder with model dimension $d_{model} = 64$, four attention heads, two encoder layers, and feed-forward dimension $d_{ff} = 96$. The reported model contains 74,281 trainable parameters, against a 75,000-parameter budget. This corresponds to 99.04% utilization of the stated budget. The architecture is deliberately compact; the goal is not to maximize language-modeling capacity but to provide a small contextual predictor that can be coupled to an embedded communication receiver.

For a candidate missing sequence $\hat{Y}_m$, exact reconstruction accuracy is defined as the fraction of evaluated missing segments that exactly match the transmitted segment. Task or field accuracy is computed separately for application-relevant fields. This distinction is important because a semantic receiver can preserve a task-critical state while failing to reproduce every character of the original message.

C. Semantic Failure Example

Consider a payload containing

TEMP=27.4, HUM=61, MOTOR=ON.

A reconstruction such as

TEMP=21.4, HUM=61, MOTOR=ON

may remain syntactically valid while being semantically wrong. A CRC cannot certify such a reconstruction because the CRC is calculated over the original transmitted bits. Consequently, semantic plausibility and exact correctness must be separated in the recovery decision.

VI. CONFIDENCE-GATED FALLBACK PROTOCOL

A. Confidence Function

The source defines the CGFP score as

$$C = C_{token} C_{phys} Penalty_{burst}(span) S_{syntax}, \quad (10)$$

where C_{token} is the BASR token-level confidence, C_{phys} represents confidence associated with the physical observation, $Penalty_{burst}$ is a nonincreasing function of burst span, and S_{syntax} is a hard structural-validity indicator.

The structural factor is deliberately binary. If the reconstructed payload violates a mandatory format or an admissible field constraint, $S_{syntax} = 0$ and the result is rejected regardless of the learned model confidence. This is a conservative design choice: semantic confidence is treated as a routing signal, not as proof of exact correctness.

The operating decision is

$$D = \begin{cases} \text{Accept semantic reconstruction,} & C \geq \tau, \\ \text{Selective exact fallback,} & C < \tau. \end{cases} \quad (11)$$

The calibrated threshold reported in the source is $\tau^* = 0.80$.

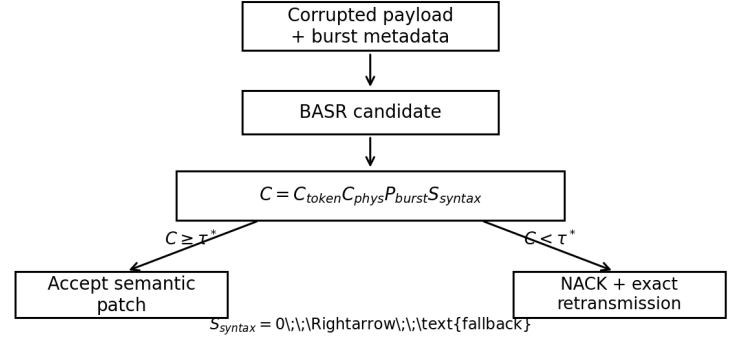


Fig. 3. CGFP decision logic. Structural invalidity forces fallback; otherwise the calibrated confidence threshold selects between semantic acceptance and exact recovery.

B. Burst-Severity Penalty

The physical burst information is incorporated through a nonlinear nonincreasing penalty. The source records the penalty parameters as $\alpha = 0.50$ and $\gamma = 1.15$. The purpose of the term is qualitative and operational: as the fraction or span of missing content increases, confidence in a contextual reconstruction should not remain unchanged. The exact calibration is tied to the studied payloads and measured burst distribution.

C. Selective Exact Fallback

When $C < \tau^*$, the receiver issues a selective negative acknowledgment (NACK) and requests the missing segment or affected frame through the exact-recovery path. The recovered content replaces the semantic candidate. This design means that semantic inference is an optimization opportunity rather than a mandatory reliability mechanism.

D. Illustrative Decision

For an illustrative frame with $C_{token} = 0.93$, $C_{phys} = 0.90$, $Penalty_{burst} = 0.95$, and valid syntax, the resulting score is

$$C = 0.93 \times 0.90 \times 0.95 \approx 0.795. \quad (12)$$

Since this value is below $\tau^* = 0.80$, CGFP rejects the semantic candidate and invokes exact fallback. The example demonstrates the conservative operating point: a candidate can appear strong while still being rejected when its combined confidence is marginal. If a candidate violates a hard structural rule, $S_{syntax} = 0$ and fallback is immediate.

VII. EXPERIMENTAL METHODOLOGY

A. Experimental Stages

The reported study is organized into four stages. First, clean-link transmissions establish the physical-layer baseline and CRC behavior. Second, controlled servo-driven occlusion generates the empirical burst dataset. Third, BASR is evaluated using corruption patterns derived from the measured burst distribution, while CGFP is calibrated and tested on disjoint burst sets. Fourth, the integrated SemLiFi pipeline is compared

with conventional recovery mechanisms under simulated and live optical conditions.

The independent variables are blockage duration/profile, payload structure, link configuration, and confidence threshold. The dependent variables include exact reconstruction accuracy, task/field accuracy, recovery latency, UFER, retransmission overhead, false acceptance, and false rejection.

B. Baselines

Four recovery conditions are considered:

- 1) **No Recovery:** the decoded payload is delivered without correction; this serves as a qualitative lower-level reference.
- 2) **RS-FEC:** RS(33,25) is used for exact symbol correction, with $t = 4$ symbol errors as the selected correction capability.
- 3) **Stop-and-Wait ARQ:** the receiver requests retransmission of the affected frame when exact recovery is required.
- 4) **SemLiFi + CGFP:** BASR generates a semantic candidate and CGFP either accepts it or invokes selective exact recovery.

The source specifies that comparisons should use the same payload population and comparable burst conditions. Where a latency value is a clean-link reference rather than a burst-recovery latency, it is explicitly labeled as such; this distinction is important for avoiding an invalid baseline comparison.

C. Evaluation Metrics

Exact reconstruction accuracy is

$$Acc_{exact} = \frac{N_{exact}}{N_{evaluated}}. \quad (13)$$

False acceptance rate is

$$FAR = \frac{N_{incorrect\ accepted}}{N_{accepted}}. \quad (14)$$

False rejection rate is

$$FRR = \frac{N_{correct\ rejected}}{N_{correct\ opportunities}}. \quad (15)$$

Recovery latency is

$$T_{rec} = T_{output} - T_{loss}, \quad (16)$$

and total recovery time is decomposed as

$$T_{total} = T_{communication} + T_{detection} + T_{inference} + T_{fallback}. \quad (17)$$

The end-to-end undetected frame error rate (UFER) is

$$UFER = \frac{N_{incorrect\ final}}{N_{evaluated\ frames}}. \quad (18)$$

The source treats a 2% UFER value as the preregistered acceptability bound; because the current manuscript record does not contain the preregistration document itself, this status must be independently verified before submission.

D. Statistical Treatment

The reported BASR exact accuracy and SemLiFi UFER values are expressed as mean $\pm$ standard deviation across five simulation seeds: 42, 101, 2024, 7, and 99. The variation therefore represents sensitivity to sampled frames and burst events under a fixed model and pipeline configuration, rather than variance from five independent model-training runs. The CGFP calibration/test split is 664/641 bursts and is described as disjoint at the event level.

VIII. ANALYTICAL INTERPRETATION OF BURST RECOVERY

A. Burst Length and Coding Capability

The physical impairment and the coding configuration can be compared at the symbol level. For a bit period T_b , a burst of duration T_B affects approximately $N_B \approx T_B/T_b$ bits before accounting for frame boundaries, Manchester expansion, and the distinction between bits and coded symbols. With the reported $T_b = 1000 \mu s$, even the shortest P1 obstruction interval of approximately 15 ms can affect a substantial contiguous region of the transmitted stream. The long P2 intervals can affect a much larger region. The exact number of corrupted bytes depends on where the obstruction begins and ends relative to the frame and on the Manchester decoding boundary.

The selected RS(33,25) configuration has 8 parity symbols and a nominal correction capability of $t = 4$ symbol errors under the evaluated decoding configuration. The physical experiment reports approximately 10–25 contiguous corrupted bytes during obstruction. This does not mean that every event necessarily produces exactly 10–25 uncorrectable RS symbols, because frame boundaries, symbol definitions, and the decoder's erasure/error handling affect the effective correction outcome. It does, however, explain why a finite correction window can be exceeded by the measured physical impairment. The reported 37.5% recovery rate is therefore interpreted as an empirical result for the implemented configuration, not as a theoretical claim that RS codes cannot handle optical burst errors.

B. Why the Semantic Path Can Reduce Retransmission

Suppose an affected frame contains a structured payload in which some fields are predictable from surviving context. An exact recovery system treats the missing region as unavailable information and requests it again. A semantic recovery system instead computes a candidate from Y^- , Y^+ , and B . If the candidate passes the CGFP gate, the receiver can deliver the candidate without using the exact retransmission path. The theoretical retransmission-saving fraction for a set of N frames can be written as

$$R_{save} = 1 - \frac{M}{N}, \quad (19)$$

where M is the number of frames that still require exact recovery. The source reports 27.15% retransmissions avoided at the selected operating point in simulation. Because the live 15-frame run rejected every candidate, R_{save} should not be interpreted as a measured hardware quantity.

The trade-off is asymmetric. A false rejection causes an unnecessary exact recovery operation and therefore costs latency or channel use. A false acceptance can silently deliver

an incorrect value. CGFP is designed around this asymmetry. The multiplicative confidence rule makes a low physical confidence or a hard syntax failure sufficient to suppress semantic delivery even when the learned model is individually confident.

C. Confidence Calibration as a Communication-Layer Function

The confidence threshold should not be viewed as a generic machine-learning hyperparameter. In SemLiFi, it determines a communication-layer routing decision. The calibration data therefore need to represent the physical burst distribution and the payload schemas for which the system will operate. If the threshold is calibrated only on clean or synthetically corrupted data, it may not represent the behavior of the receiver under real optical occlusion.

The reported 664/641 burst split provides a basis for a disjoint calibration/test design. The calibration set should be used to select τ and the burst-penalty parameters, while the test set should remain untouched until the final evaluation. This separation is especially important when the objective is a safety-oriented metric such as UFER. Threshold selection after observing the test errors would produce an optimistic estimate.

A complete threshold study should report at least the acceptance rate, exact precision, FAR, FRR, UFER, retransmission-avoidance rate, and recovery latency at each candidate threshold. The present source reports the selected operating point but not the complete sweep. Accordingly, the current paper treats $\tau^* = 0.80$ as a reported operating point and identifies the underlying sweep as a reproducibility item rather than reconstructing missing measurements.

D. Task-Level Accuracy and Safety-Critical Fields

Exact reconstruction and task-level recovery answer different questions. Exact reconstruction asks whether the missing representation is identical to the transmitted representation. Task-level accuracy asks whether a predefined field needed by the application is correct. The reported BASR results contain a substantial difference between these two metrics: exact accuracy is $42.63 \pm 1.47\%$, whereas the evaluated motor-state field reaches 100.0% accuracy.

This difference is useful only if the task field definition is explicit. For example, a motor state encoded as `MOTOR=ON` can be treated as correct when the recovered state is exactly `ON`, even if another non-critical field differs. The 100.0% value should therefore not be interpreted as 100.0% exact payload recovery. The denominator and event definition used for this metric must be retained with the raw evaluation data.

For safety-oriented telemetry, the safest policy may be to define a field-specific acceptance rule. A reconstruction that changes an actuation state could be forced to exact fallback even when other fields have high confidence. The current CGFP implementation provides a general structural-validity factor, but the source does not report a completed field-specific risk model. Such a model is a natural extension of the framework.

E. Latency Decomposition

The reported end-to-end latency can be understood as the sum of communication, burst detection, inference, validation, and fallback components. In the semantic-acceptance case, the fallback term is absent; in the rejection case, the exact-recovery term becomes dominant. This makes the system's performance dependent on the distribution of confidence outcomes rather than only on the raw inference time.

The reported 2.79 ms mean BASR inference time is therefore not directly comparable with the 790.7 ms end-to-end continuous-burst latency. The former measures the model computation on an edge CPU, while the latter includes the full communication and recovery path. The difference in scale also indicates why optimizing the physical and retransmission components can be more important than shaving small amounts from the neural inference time.

The live hardware latency range of 636.6–792.9 ms should similarly be interpreted with its recovery behavior. Because all 15 frames triggered fallback, the observed latency includes exact-recovery actions rather than semantic-patch acceptance. The result cannot establish what the latency would be for a live frame that passes CGFP.

IX. EXPERIMENTAL REPRODUCIBILITY AND STATISTICAL RIGOR

A. Separation of Training, Calibration, and Testing

A defensible experimental pipeline should keep three kinds of information separate. The BASR training data determine model parameters. The CGFP calibration data determine the threshold and confidence mapping. The final test data estimate the reported performance. If bursts from the same physical event are split across these sets, leakage can occur because adjacent observations may share nearly identical optical conditions.

The supplied record explicitly describes the 664/641 CGFP calibration/test split as disjoint at the event level. This is a useful property. For the final paper, the authors should also preserve the event identifier, physical profile, frame identifier, payload instance, and random seed for each sample so that the split can be audited.

B. Simulation Versus Hardware Evidence

The study combines physical measurements with simulation. This is appropriate for a proof-of-concept system because the hardware campaign establishes that the impairment exists and the simulation permits larger controlled comparisons. However, the two evidence types answer different questions.

The 56-event physical campaign measures the burst phenomenon. The BASR and CGFP simulation evaluates the recovery strategy over a larger set of derived conditions. The 15-frame hardware run tests the integrated decision behavior under actual optical conditions. Because the hardware run accepts no semantic patches, it cannot validate the simulation's retransmission-avoidance efficiency. This distinction should remain explicit in every abstract, table, figure, and conclusion.

C. Meaning of Mean Plus or Minus Standard Deviation

The source reports $42.63 \pm 1.47\%$ exact accuracy and $1.00 \pm 0.13\%$ UFER across five simulation seeds. The reported statistical treatment states that these seeds vary sampled frames and burst events under a fixed model and pipeline configuration. Thus, the standard deviation should not be described as model-training uncertainty. To estimate training variability, independent training runs with different initialization seeds would be required.

Likewise, a five-seed simulation does not establish a confidence interval for deployment performance. The variance is useful as a sensitivity measure within the evaluated simulation design, but the live hardware population is currently much smaller. The paper therefore avoids attaching a population-level statistical guarantee to the reported 15-frame hardware result.

D. Avoiding Leakage Through Payload Templates

Structured telemetry can contain repeated templates. If the same payload instance or nearly identical field sequence appears in both training and testing, exact reconstruction accuracy may be inflated because the model can exploit memorized structure. A stronger future protocol should split by payload template or device identifier in addition to splitting by burst event. This would test whether BASR learns general structural relationships rather than simply memorizing recurring messages.

E. Ablation Protocol Required for Causal Claims

The central novelty depends on the interaction between burst awareness and confidence gating. A rigorous ablation should therefore include at least the following configurations: random-mask BASR without physical burst features; burst-aware BASR without burst features in CGFP; CGFP without the physical confidence factor; CGFP without the burst penalty; CGFP without the syntax factor; and the full system. Each configuration should be evaluated on the same held-out event set.

The current source identifies this experiment as missing. Consequently, the paper does not claim that every individual factor is necessary based on the current measurements. Instead, it reports the performance of the integrated configuration and presents the ablation as a required next experiment.

F. Recommended Reporting Matrix

For each operating condition, a final dataset should contain the physical burst profile, duration, estimated affected symbols, frame length, BASR confidence, structural-validity result, semantic decision, exact-fallback decision, final correctness, task-field correctness, inference latency, and total recovery latency. This event-level record would allow reviewers to reproduce aggregate metrics and inspect false acceptances directly.

TABLE III
RECOMMENDED EVENT-LEVEL REPORTING FIELDS

Field	Purpose
Event/profile ID	Physical-condition traceability
T_B and affected symbols	Burst severity
Payload/template ID	Leakage control
C_{token}, C_{phys}	Confidence audit
$Penalty_{burst}$	Physical penalty audit
S_{syntax}	Structural safety check
CGFP decision	Accept/reject routing
Exact correctness	Bit/segment ground truth
Task-field correctness	Application utility
Fallback latency	Exact recovery cost
Final UFER flag	End-to-end reliability

X. ENGINEERING DESIGN TRADEOFFS

A. ESP32 Versus More Powerful Edge Hardware

The ESP32 is appropriate for the proof-of-concept because the transmitter and receiver require GPIO timing, ADC acquisition, Manchester decoding, burst detection, and communication control in addition to the semantic inference path. A more powerful processor could support larger models or higher rates, but it would also make the prototype less representative of a low-cost embedded edge implementation. The present 74,281-parameter model is intentionally constrained to keep the inference component small.

This choice does not imply that every Transformer architecture can run on an ESP32. The reported 2.79 ms value is an edge CPU measurement associated with the evaluated BASR implementation, and the complete deployment footprint depends on the runtime, memory allocator, numerical representation, and firmware integration. These details should be measured explicitly before claiming microcontroller-level deployment on a specific board.

B. Why Manchester Coding Is Retained

Manchester coding introduces transition redundancy and therefore reduces raw payload efficiency compared with a simple binary level scheme. It nevertheless provides a useful timing structure for a small experimental testbed. The receiver can use mid-bit sampling and transition checks to distinguish valid symbols from ambiguous optical observations. Since the study is focused on recovery logic rather than maximum throughput, this trade-off is acceptable within the tested configuration.

C. CRC and Semantic Recovery Serve Different Roles

CRC-8 validates bit-exact frames. BASR reconstructs missing content when bit-exact information is unavailable. These functions should not be conflated. A CRC failure identifies that the received frame is not an exact copy; it does not tell the receiver what the missing value should be. Conversely, a semantic candidate that passes a syntax check does not satisfy the CRC of the original frame because the original missing bits remain unknown.

The hybrid architecture therefore uses CRC and semantic confidence at different stages. CRC is used for exact validation, while CGFP decides whether a semantic candidate is

sufficiently trustworthy to avoid exact recovery. When exact fallback succeeds, the recovered frame can be validated with CRC.

D. Why the Servo Flap Matters

A manually generated blockage can vary substantially in duration, angle, and position. The servo flap provides repeatability and allows the study to generate three broad duration profiles. This does not make the resulting distribution representative of every human or environmental obstruction, but it provides a controlled experimental variable. Reproducibility is especially important when comparing model training on burst-aware masks against training on random masks.

E. Safety-Conservative Operating Point

The multiplicative confidence rule has a conservative interpretation. If any factor is small, the final confidence is reduced. A hard syntax failure sets the score to zero. The system therefore prefers false rejection over false acceptance when the confidence evidence is weak. This is appropriate for the stated safety concern but can reduce the number of accepted semantic patches.

The current live result illustrates this trade-off. Mean confidence is approximately 0.46, well below the 0.80 threshold, and all 15 frames fall back to exact recovery. The gate therefore sacrifices retransmission avoidance to maintain conservative behavior under those hardware conditions. Whether a different threshold would provide a better operating point requires the missing threshold-sweep data and a larger hardware test set.

XI. ALGORITHMIC IMPLEMENTATION AND OPERATING ENVELOPE

A. Receiver-Side Procedure

The receiver-side procedure can be expressed as a sequence of deterministic communication operations followed by conditional semantic inference. First, the ESP32 acquires the optical waveform and applies the calibrated threshold. Second, Manchester symbols are decoded and the frame delimiter, length, sequence identifier, and CRC fields are parsed. Third, a contiguous unavailable region is identified from missing symbols, transition violations, or frame-level integrity failure. Fourth, the receiver estimates the burst duration and affected-symbol count and extracts the surviving context.

The semantic stage is entered only after the physical processing has established a candidate loss interval. BASR receives the surviving prefix, suffix, burst metadata, and payload structure and returns a candidate missing segment with token-level confidence. CGFP then evaluates structural validity and computes the combined score. If the score is below the threshold, the receiver does not deliver the semantic candidate. Instead, it requests exact recovery. The final payload is then subjected to the exact integrity mechanism available to the receiver.

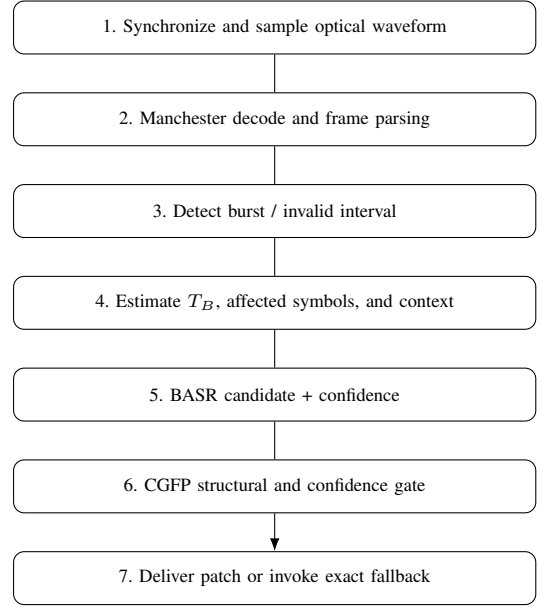


Fig. 4. Receiver-side processing sequence used to separate physical decoding, burst estimation, semantic inference, and exact fallback.

B. Pseudocode

The recovery logic can be summarized as follows:

- 1) initialize the optical receiver and calibrated threshold;
- 2) decode Manchester symbols and parse the frame;
- 3) if the frame is clean and CRC-valid, deliver it without semantic inference;
- 4) otherwise, identify the unavailable interval and estimate T_B and N_B ;
- 5) extract surviving prefix/suffix context and generate a BASR reconstruction;
- 6) compute $C = C_{token}C_{phys}Penalty_{burst}S_{syntax}$;
- 7) if $S_{syntax} = 0$ or $C < \tau^*$, issue NACK and recover exactly;
- 8) otherwise, deliver the semantic candidate and log the decision;
- 9) record final correctness, field correctness, latency, and the recovery path.

This procedure has an important operational property: a clean frame bypasses the semantic model. Semantic inference is used only when the physical layer has evidence of a loss that cannot already be resolved by the normal integrity path. This reduces unnecessary inference and prevents the model from modifying error-free traffic.

C. Operating Envelope

The demonstrated operating envelope is defined by the physical and computational parameters reported in the study: 1000 bps raw signaling, structured telemetry payloads, ESP32 transmitter and receiver nodes, BPW34 detection, LM358 TIA, Manchester coding, and servo-driven blockage. Within this envelope, the system has evidence from both simulation and a small live experiment. Outside this envelope, the evidence is incomplete.

In particular, the study does not establish performance for arbitrary distances, high-speed mobility, strong ambient

TABLE IV
CLAIM-TO-EVIDENCE MAPPING

Claim	Evidence available in current record
Clean optical link works	449.0 ms mean; zero bit errors; 100% CRC
Physical burst loss exists	56 empirical occlusion events; 10–25 contiguous bytes
BASR reconstructs structured data	$42.63 \pm 1.47\%$ exact accuracy
Task field can remain correct	100.0% reported motor-state accuracy
CGFP limits accepted errors	97.13% patch precision; 0.78% patch error
Simulation avoids retransmissions	27.15% avoided at $\tau^* = 0.80$
Simulation end-to-end result	790.7 ms; $1.00 \pm 0.13\%$ UFER
Hardware semantic efficiency	Not established; all 15 frames rejected

sunlight, multipath reflections, multiple simultaneous optical users, or high-rate LiFi modulation. The model also has not been shown to generalize across arbitrary payload schemas. These boundaries should be stated explicitly because a journal reviewer can otherwise interpret the semantic reconstruction result as a general-purpose LiFi reliability mechanism.

D. Expected Behavior With Increasing Burst Severity

As burst severity increases, the amount of surviving context decreases and the confidence penalty increases. The intended behavior is therefore a gradual shift from semantic acceptance toward exact fallback. At low severity, BASR has more context and may produce a high-confidence candidate. At intermediate severity, the confidence gate should accept only candidates with strong structural and contextual evidence. At severe loss, the system should increasingly reject semantic predictions and use exact recovery.

This behavior is preferable to a fixed semantic-only policy because the physical channel supplies information about the reliability of the reconstruction. It also provides a direct experimental hypothesis: acceptance rate should generally decrease as normalized burst span increases, while false acceptance should remain bounded if the calibration is valid. The current supplied results do not provide a complete acceptance-versus-span curve, so this relationship remains a design expectation rather than a reported quantitative law.

E. Reviewer-Focused Evidence Matrix

A useful way to interpret the evidence is to map each major claim to the experiment that supports it. The physical-link claim is supported by clean-link measurements and the 56-event occlusion campaign. The BASR capability claim is supported by the reported exact and field-level accuracy values. The CGFP safety claim is supported by accepted-patch precision, patch-error rate, and the 15-frame live rejection behavior. The retransmission-avoidance claim is supported only by simulation because the live run accepted no patches. This separation prevents the strongest simulation result from being inadvertently presented as a hardware result.

The evidence matrix is also a useful guide for revision. A stronger final paper should add a larger live dataset, the full

TABLE V
CLEAN-LINK PHYSICAL VALIDATION

Metric	Reported value
Mean transit latency	449.0 ms
Latency range	443–455 ms
Bit errors	0
CRC-8 validation	100%

threshold sweep, and the ablation results before making causal or generalization claims. The current version deliberately keeps those limitations visible rather than filling missing evidence with inferred values.

XII. RESULTS

A. Clean-Link Physical Validation

The clean-link experiment reports a mean transit latency of 449.0 ms, with a range of 443–455 ms, zero bit errors, and 100% CRC-8 validation. This baseline is important because semantic recovery should not be used to compensate for a receiver that is already unreliable in the absence of occlusion.

B. Empirical Burst Behavior

The physical campaign contains 56 occlusion events. The three profiles cover short, long, and mixed/randomized servo-driven blockages. The source reports approximately 10–25 contiguous corrupted bytes during physical obstruction. This observation is central to the motivation for burst-aware recovery: the physical corruption interval can be larger than the correction window of the selected RS-FEC configuration.

Because the supplied experimental record does not provide a complete numerical table of mean, median, and maximum duration for every P1–P3 profile, those values are not reconstructed here. Instead, the reported ranges and total event count are retained. This is deliberate: filling missing profile statistics from inference would compromise the reproducibility and accuracy of the paper.

C. BASR Reconstruction Performance

The BASR model contains 74,281 trainable parameters. The reported exact reconstruction accuracy is $42.63 \pm 1.47\%$. Although exact character-level recovery is below half of the evaluated missing segments, the reported motor actuation-state field accuracy is 100.0%. This demonstrates the distinction between exact source reproduction and task-relevant field preservation.

The model’s mean edge CPU inference latency is reported as 2.79 ms. The parameter count corresponds to an approximate 290 KB footprint when stored as 32-bit floating-point weights and approximately 73 KB under an idealized 8-bit representation. The latter is a storage estimate rather than a measurement of the complete deployed binary or runtime memory footprint.

TABLE VI
REPORTED BASR PERFORMANCE

Metric	Value
Trainable parameters	74,281
Exact reconstruction accuracy	42.63 $\pm$ 1.47%
Motor-state accuracy	100.0%
Mean edge CPU inference latency	2.79 ms
Parameter budget	75,000

TABLE VII
REPORTED END-TO-END RESULTS

Method	Reported result
RS-FEC	37.5% delivery; 62.5% UFER; 449.0 ms is clean-link reference
Stop-and-Wait ARQ	97.1% delivery; 1560.0 ms; 0.0% UFER
SemLiFi, 100% burst simulation	100.0% delivery; 790.7 ms; 1.00 $\pm$ 0.13% UFER
SemLiFi, mixed 40% burst	100.0% delivery; 592.5–595.5 ms
SemLiFi, live hardware ($N = 15$)	100.0% delivery; 636.6–792.9 ms; 0.00% UFER

D. Confidence-Gating Operating Point

The reported CGFP threshold is $\tau^* = 0.80$. At this operating point, approximately 27.15% of retransmissions are avoided through accepted semantic patches in the evaluated simulation. Accepted patches have 97.13% precision, and the reported patch-error rate is 0.78%. These values are directly relevant to the design objective because the main safety concern is not merely rejection of a weak candidate but accidental acceptance of an incorrect one.

The complete multi-point threshold sweep underlying the selected threshold is not present in the supplied experimental record. Therefore, the paper does not invent intermediate values at thresholds such as 0.50, 0.60, 0.70, or 0.90. The threshold should be regarded as a reported operating point requiring release of the underlying sweep for full reproducibility.

E. End-to-End Comparison

The source reports the following end-to-end values. RS-FEC recovers 37.5% of the real occlusion bursts and consequently leaves a 62.5% UFER in that evaluated burst set. The 449.0 ms value associated with the RS-FEC row is a clean-link reference, not a claim that RS-FEC burst recovery itself completes in 449.0 ms. Stop-and-Wait ARQ reports 97.1% delivery at 1560.0 ms mean latency and 0.0% UFER.

Under continuous-burst simulation, SemLiFi reports 100.0% delivery at 790.7 ms mean latency with UFER 1.00 $\pm$ 0.13%. Under the mixed condition in which 40% of frames experience bursts, the reported latency is 592.5–595.5 ms with 100.0% delivery. The source does not provide a comparable UFER value for this mixed condition, so none is inserted.

F. Continuous-Burst Simulation

The continuous-burst simulation is the principal quantitative comparison between the hybrid architecture and Stop-and-Wait ARQ. SemLiFi reaches 790.7 ms mean latency compared with

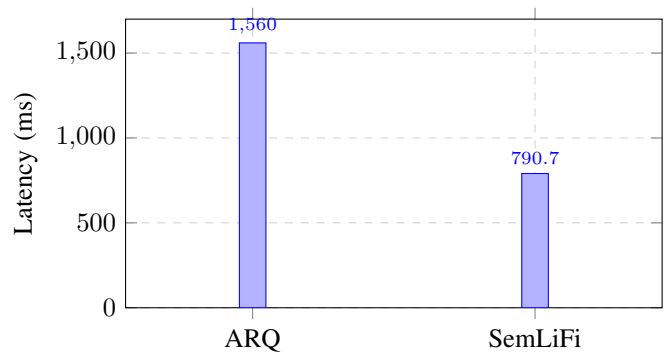


Fig. 5. Reported mean latency under continuous-burst simulation for Stop-and-Wait ARQ and SemLiFi. The comparison uses the source-reported operating conditions.

1560.0 ms for ARQ, while both reported conditions achieve 100% delivery. The reduction is

$$1 - \frac{790.7}{1560.0} \approx 49.3\%. \quad (20)$$

The simulated SemLiFi UFER of 1.00 $\pm$ 0.13% is below the reported 2% bound. However, this is a simulation result using the stated five seeds and evaluated burst distribution. It should not be interpreted as a universal LiFi reliability guarantee.

G. Mixed-Burst Simulation

When 40% of frames experience bursts, SemLiFi reports 100% delivery and a latency range of 592.5–595.5 ms. This is lower than the continuous-burst condition because fewer frames require recovery actions. The result illustrates the expected dependence of total recovery cost on burst prevalence, but it should not be used to infer a continuous functional latency curve because the source reports only this mixed operating point.

H. Live Hardware Evaluation

The live hardware evaluation contains 15 frames. It reports 100% delivery, a latency range of 636.6–792.9 ms, and 0.00% UFER. Importantly, all low-confidence semantic reconstructions were rejected. The mean confidence was approximately 0.46, below $\tau^* = 0.80$, and selective NACKs were therefore used for all 15 frames.

This result is positive evidence for the safety behavior of the gate under the tested severe conditions, but it is not evidence that semantic patching is efficient on hardware. Since no live frame was accepted as a semantic patch, the 27.15% retransmission-avoidance result remains a simulation result. A larger hardware campaign is required before claiming that the simulated efficiency transfers to physical operation.

XIII. DISCUSSION

A. Semantic Usefulness Versus Exact Correctness

The BASR result demonstrates why semantic usefulness and exact correctness should be reported separately. An exact reconstruction accuracy of 42.63% would be insufficient if the objective were bit-perfect replacement of every missing

segment. The reported 100.0% motor-state accuracy, however, indicates that the evaluated application field remained correct despite imperfect character-level reconstruction. This does not make the semantic output safe by itself; an incorrect value in another field can still be operationally harmful. CGFP therefore provides the missing decision layer.

B. Why Burst Awareness Matters

The physical impairment is not an arbitrary mask. It is a contiguous loss interval produced by an obstruction. Burst duration determines how much context survives and how much of the payload is unavailable. Including burst metadata in BASR and in the confidence calculation therefore connects the physical and semantic layers. The approach is intended to be more physically grounded than training solely on independent random token masks.

C. Why Exact Fallback Remains Necessary

The central design principle is that semantic reconstruction is an optional shortcut to exact recovery, not a replacement for exact recovery. If the semantic candidate is sufficiently trustworthy under the calibrated criterion, the system can avoid retransmission. If confidence is low or structural validation fails, exact recovery remains available. This architecture is particularly relevant when the cost of a false semantic acceptance is larger than the cost of a false rejection.

D. Interpretation of the Latency Result

The approximately 49.3% reduction relative to the reported ARQ latency is meaningful only under the continuous-burst simulation conditions stated in the source. The result does not establish a universal latency advantage because latency depends on burst duration, retransmission protocol, frame length, processing hardware, and the availability of the return channel. The 2.79 ms BASR inference time is comparatively small relative to the reported end-to-end latency, but it is not the only computation or communication cost.

E. Practical Deployment Considerations

A physical deployment would need to account for ambient illumination, optical alignment, dimming behavior, LED driver dynamics, receiver saturation, and mobility. The approximate 50-count receiver threshold was calibrated under the reported laboratory conditions. A moving receiver or a change in ambient light could shift the illuminated and dark distributions and therefore alter the effective confidence term.

The 1000 bps demonstration rate also limits the scope of the current evaluation. A higher-rate implementation could change the relationship between physical burst duration and affected bytes. In particular, at a higher rate the same physical obstruction could erase more symbols, increasing the burden on both BASR and exact fallback.

F. Energy Considerations

The source does not report direct energy measurements. It is reasonable as an engineering hypothesis that avoiding retransmissions can reduce repeated transmitter and receiver activity, but this has not been quantified in the study. Therefore, energy savings are treated as a future measurement rather than a demonstrated result.

G. Broader Optical Applicability

The architectural pattern could be investigated for other optical channels experiencing contiguous loss, including free-space and underwater optical links. However, the burst distribution, physical confidence calibration, and structural payload constraints would have to be re-characterized for each channel. The current results should not be transferred directly to those settings.

XIV. LIMITATIONS AND THREATS TO VALIDITY

A. Physical Realism of Occlusion

The servo flap provides repeatable blockage but cannot reproduce every real obstruction. Human motion, partial opacity, object shape, angle of incidence, receiver orientation, and ambient illumination can produce different burst statistics. The three measured profiles are therefore a controlled starting point rather than an exhaustive model of physical LiFi blockage.

B. Payload Generality

BASR depends on structured or partially predictable telemetry. It should not be generalized to arbitrary encrypted or structureless binary data. Different payload schemas may require different model training and structural-validity rules.

C. Live-Hardware Sample Size

The live experiment contains only 15 frames. More importantly, every frame was rejected by CGFP. The live run therefore supports the rejection/fallback safety behavior but does not establish a hardware patch-acceptance rate or hardware retransmission-avoidance efficiency.

D. Missing Component Ablations

The available experimental record does not contain completed ablation measurements separating burst-aware training from random masking or isolating the contribution of individual CGFP factors. As a result, the current data cannot prove that each component independently causes the reported end-to-end improvement. This should be addressed before making stronger causal claims.

E. Threshold Calibration

The reported threshold $\tau^* = 0.80$ and the burst-penalty parameters are specific to the evaluated telemetry and burst distribution. The complete threshold sweep is not included in the supplied record. The operating point should therefore be reproduced from the original calibration logs before final submission.

XIV. LIMITATIONS AND THREATS TO VALIDITY

A. Physical Realism of Occlusion

The servo flap provides repeatable blockage but cannot reproduce every real obstruction. Human motion, partial opacity, object shape, angle of incidence, receiver orientation, and ambient illumination can produce different burst statistics. The three measured profiles should therefore be interpreted as a controlled experimental starting point rather than an exhaustive model of physical LiFi blockage. The controlled actuator is valuable for repeatability, but it does not establish that the measured distribution is representative of all indoor optical environments.

B. Payload Generality

BASR depends on structured or partially predictable telemetry. The evaluated examples contain explicit field names, delimiters, numeric ranges, and state labels. These properties provide contextual constraints that are not available for arbitrary encrypted or structureless binary payloads. Different schemas would require their own training distribution, parsing rules, and structural-validity constraints. The reported accuracy should therefore be scoped to the structured-telemetry setting used in the study.

C. Live-Hardware Sample Size

The live experiment contains only 15 frames, and every low-confidence reconstruction was rejected by CGFP. The experiment therefore provides evidence about conservative fallback behavior rather than about the hardware acceptance rate of semantic patches. In particular, the reported 27.15% retransmission avoidance remains a simulation result and should not be presented as a measured hardware efficiency.

D. Missing Component Ablations

The supplied record does not contain completed ablation measurements that separately isolate burst-aware training, the physical-confidence factor, the burst penalty, and the structural-validity term. Consequently, the current evidence supports the integrated system result but does not establish that each individual component is necessary for the observed performance. A controlled ablation on the same held-out event set is required before stronger causal claims are made.

E. Threshold Calibration

The reported operating threshold is $\tau^* = 0.80$, with burst-penalty parameters $\alpha = 0.50$ and $\gamma = 1.15$. The complete threshold sweep is not included in the supplied record. Threshold selection should therefore be reproduced from the original calibration logs, and the final paper should report acceptance rate, precision, FAR, FRR, UFER, retransmission avoidance, and recovery latency across the candidate thresholds.

F. UFER Acceptance Bound

The manuscript describes a 2% UFER acceptance bound, but the supplied record does not include the preregistration document. The status of that bound should be independently verified before submission. If no formal preregistration exists, the wording should identify the criterion as an evaluation threshold rather than a preregistered commitment.

G. Bibliographic and Author Verification

The final submission should preserve the verified conference-paper metadata for the Nennouche et al. citation [11] and should be checked by all authors for author order, affiliations, and institutional e-mail addresses. These checks are editorial requirements and are not experimental evidence.

XV. REPRODUCIBILITY AND IMPLEMENTATION DETAILS

A. Reproduction Sequence

A reproducible run should begin with hardware assembly and receiver calibration, followed by clean-link verification. The three servo-driven obstruction profiles should then be executed while preserving event timing, burst duration, affected symbols, frame identifiers, and local signal measurements. BASR masks should be generated from the

measured burst distribution rather than from independent random masking alone. The CGFP calibration/test split should remain disjoint at the event level, and the reported simulation seeds should be retained.

B. Implementation Organization

Component	Role
Sender.ino	OOK transmit firmware
Receiver.ino	OOK receive firmware
servo_rig.ino	Occlusion controller
burst_logger.py	Burst logging
data/burst_events.jsonl	Raw burst-event log
rs_fec.py	Reed–Solomon baseline
arq_baseline.py	ARQ baseline
basr/model.py	BASR architecture
basr/train.py	BASR training
cgfp/confidence.py	Confidence engine
cgfp/patcher.py	Decision/patch logic
integration/full_pipeline.py	Unified pipeline

C. Reproducibility Metadata

The reported simulation seeds are 42, 101, 2024, 7, and 99. A complete reproducibility package should additionally identify the software versions, operating system, ESP32 firmware revision, compiler/toolchain, dataset hashes, model checkpoint hash, and the full threshold sweep. These items are not fully specified in the supplied manuscript and should be taken from the development records rather than inferred.

D. Event-Level Logging

For each obstruction event, the retained record should include the profile identifier, start and end times, burst duration, estimated affected symbols, frame identifier, payload or template identifier, BASR confidence, physical-confidence term, burst penalty, syntax-validity flag, CGFP decision, final correctness, and recovery latency. Keeping these fields together allows aggregate results to be recomputed directly from event records and makes false acceptances auditable.

E. Experimental Separation

Clean-link validation, physical burst characterization, BASR evaluation, CGFP calibration, and integrated end-to-end evaluation should remain distinct experimental stages. A clean frame should bypass semantic inference, while a burst event should be processed only after the physical receiver establishes an invalid or unavailable interval. This separation prevents semantic processing from altering error-free traffic and helps isolate communication, inference, and fallback costs.

XVI. FUTURE WORK AND EXPERIMENTAL EXTENSIONS

A. Larger Hardware Campaign

The most important next experiment is a larger live-hardware evaluation that deliberately covers moderate burst severities. The current 15-frame run rejected every semantic reconstruction, so a broader severity range is needed to obtain enough accepted patches to estimate hardware patch precision and patch error. The experiment should preserve event-level identifiers so that physical conditions can be separated from repeated payload templates.

B. Controlled Ablation Study

BASR should be trained and evaluated under at least two masking regimes: independent random masking and masks sampled from the measured optical burst distribution. The CGFP study should then remove one confidence factor at a time while keeping the same test events. This design can distinguish the effect of physical burst awareness from the effect of confidence gating and structural validation.

C. Threshold-Sweep Publication

A complete threshold study should report acceptance rate, exact precision, FAR, FRR, UFER, retransmission avoidance, and latency over a dense candidate set. The selected $\tau^* = 0.80$ can then be justified using a criterion defined before the final test evaluation. Reporting the complete curve is particularly important because a single operating point does not reveal the reliability–latency trade-off.

D. Higher-Rate Evaluation

The prototype uses a 1000 bps raw signaling rate. Increasing the rate changes the number of symbols affected by a fixed physical obstruction duration. The resulting interaction among burst length, surviving context, BASR accuracy, inference time, and retransmission cost should therefore be measured directly rather than extrapolated from the current operating point.

E. Broader Physical Conditions

Future physical experiments should vary obstruction opacity, size, angle, speed, and receiver orientation. Ambient-light conditions and source–receiver separation should also be varied independently. Such measurements would provide a broader empirical basis for calibrating C_{phys} and P_{burst} . Cross-domain evaluation in free-space or underwater optical links would require new burst characterization and payload-specific calibration.

F. Security-Oriented Extensions

If the architecture is considered for safety-relevant deployments, deliberate telemetry spoofing, syntactically valid false data, adversarial burst patterns, and optical manipulation should be evaluated separately. These cases are outside the current non-adversarial obstruction threat model and should not be conflated with the reported UFER measurements.

XVII. ETHICAL AND SAFETY CONSIDERATIONS

A. Semantic Substitution Risk

The principal safety concern is silent semantic substitution: a reconstructed value may be plausible and syntactically valid while differing from the transmitted value. Semantic confidence is therefore treated as a routing signal rather than as proof of correctness. The structural gate and exact fallback are designed to reduce the chance that an uncertain candidate is delivered as if it were exact.

B. Scope of the Safety Evidence

The current study does not evaluate deliberate frame forgery, malicious telemetry injection, adversarial examples, or intentional optical manipulation. The reported UFER should therefore be interpreted only within the tested physical-burst and structured-telemetry setting. The live experiment produced 0.00% UFER for 15 frames, but the small sample does not establish a population-level security or reliability guarantee.

XVIII. EXTENDED QUANTITATIVE ANALYSIS

A. Coding Capability Versus Observed Burst Length

The selected RS(33,25) configuration contains eight parity symbols and has a nominal correction capability of $t = 4$ symbol errors under the evaluated configuration. The physical campaign reports approximately 10–25 contiguous corrupted bytes during obstruction. This difference explains why a finite correction window can be exceeded in some events, although the exact mapping from corrupted bytes to decoder symbols depends on frame boundaries, Manchester coding, and the decoder treatment of erasures and errors. The reported 37.5% recovery rate is therefore an empirical result for the implemented configuration rather than a theoretical limit on Reed–Solomon coding.

B. Retransmission Avoidance

For N affected frames, if M frames still require exact recovery, the retransmission-saving fraction is $R_{\text{save}} = 1 - M/N$. At the reported operating point, the simulation avoids 27.15% of retransmissions. Because every one of the 15 live frames invokes fallback, no hardware retransmission-saving percentage is established by the current experiment.

C. Task-Level Reliability

The difference between exact reconstruction and task-level correctness is not only a reporting distinction; it changes the acceptance policy. A field that controls actuation or a safety-relevant state can be assigned a stricter rule than a non-critical descriptive field. Under the current architecture, such a policy can be implemented by extending the structural-validity factor so that selected fields force exact fallback whenever their reconstructed value is uncertain.

D. Confidence Factor Interaction

The multiplicative CGFP score makes the decision sensitive to each factor. A low physical-confidence estimate, a severe burst penalty, or a low token-level confidence can reduce the combined score below τ^* . The hard syntax factor is stronger still: $S_{\text{syntax}} = 0$ forces fallback regardless of learned confidence. This design reflects the asymmetric cost of false acceptance versus false rejection.

E. Mixed-Burst Operating Point

Under the reported mixed condition in which 40% of frames experience bursts, SemLiFi achieves 100% delivery with a latency range of 592.5–595.5 ms. The supplied record does not provide a comparable UFER value for this condition. The result therefore illustrates the dependence of total recovery cost on burst prevalence without establishing a continuous latency curve.

XVIII. EXTENDED QUANTITATIVE ANALYSIS (CONTINUED)

C. Latency Decomposition

The total recovery time can be decomposed as $T_{\text{total}} = T_{\text{communication}} + T_{\text{detection}} + T_{\text{inference}} + T_{\text{fallback}}$. The reported BASR inference time is 2.79 ms, whereas the continuous-burst end-to-end SemLiFi latency is 790.7 ms. These quantities measure different stages and should not be compared as if they were equivalent. The large difference indicates that retransmission and communication components dominate the end-to-end timing under severe burst conditions.

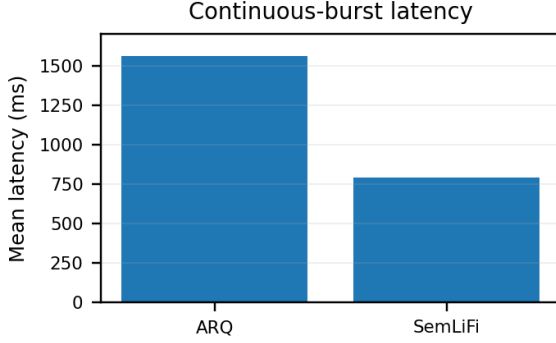


Fig. 6. Reported mean latency under the continuous-burst simulation. The 1560.0 ms ARQ value and 790.7 ms SemLiFi value are taken directly from the supplied experimental record.

D. Relative Latency Reduction

Using the reported continuous-burst values, the relative reduction with respect to the Stop-and-Wait ARQ condition is $1 - 790.7/1560.0 \approx 49.3\%$. This calculation is a derived comparison of the reported operating point. It does not establish a universal latency advantage because the result depends on burst prevalence, frame length, return-channel behavior, retransmission policy, and processing hardware.

E. BASR Exact and Task-Level Accuracy

The reported exact reconstruction accuracy is $42.63 \pm 1.47\%$, while the evaluated motor-state field reaches 100.0% accuracy. The distinction is central: exact reconstruction requires the missing segment to match the transmitted representation, whereas task-level accuracy asks whether a predefined application field is correct. The latter can remain correct even when other non-critical characters differ. The denominator and event definition for the motor-state metric should be retained with the raw evaluation data.

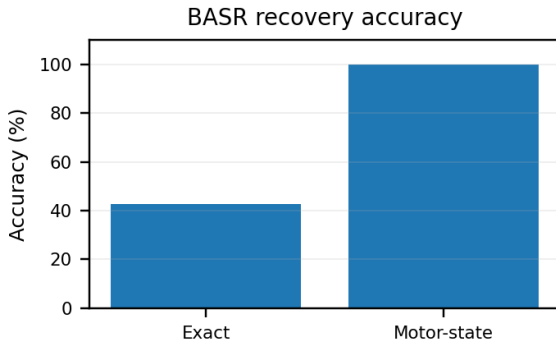


Fig. 7. Reported BASR exact reconstruction and motor-state field accuracy. The values are not interchangeable and represent different evaluation objectives.

F. Model Size and Edge Constraints

The BASR model contains 74,281 trainable parameters against a 75,000-parameter budget, corresponding to 99.04% utilization. A simple 32-bit storage calculation gives approximately 290 KB for the weights, while idealized 8-bit storage gives approximately 73 KB. These are storage estimates, not measurements of the complete runtime footprint. The reported 2.79 ms inference time is an edge-CPU measurement and

should not be interpreted as proof that the entire Transformer runtime fits within an ESP32 memory budget without additional deployment measurements.

G. CGFP Calibration/Test Separation

The reported calibration and test sets contain 664 and 641 bursts, respectively, and are described as disjoint at the event level. This separation is important because threshold selection must not use the final test outcomes. The split provides a clear basis for auditing whether the calibrated threshold was selected independently of the final performance estimate.

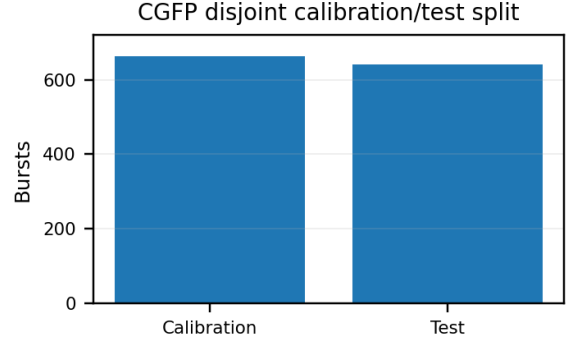


Fig. 8. Reported event-level separation of the CGFP calibration and test sets.

H. Threshold Interpretation

The selected threshold should be interpreted as a communication-layer routing point rather than as a generic machine-learning hyperparameter. A higher threshold would generally be expected to reduce semantic acceptance while increasing exact fallback, whereas a lower threshold would admit more candidates and require careful monitoring of false acceptance. Because the complete sweep is not supplied, these directional effects are design expectations rather than measured curves.

I. Evidence Boundary

The continuous-burst simulation and the live hardware run answer different questions. The simulation provides the reported latency, UFER, and retransmission-avoidance measurements under controlled seeds and burst conditions. The hardware run establishes that the physical receiver, burst detector, BASR path, and fallback gate can operate together, but its 15-frame size and universal fallback behavior do not support a hardware estimate of semantic acceptance efficiency.

XVIII. EXTENDED QUANTITATIVE ANALYSIS (CONTINUED)

H. Live-Hardware Decision Behavior

The live evaluation contains 15 frames. The mean semantic confidence is approximately 0.46, below the calibrated threshold of 0.80, and all 15 candidates were rejected. Consequently, the measured live result is 100% delivery with 0.00% UFER over the reported run, but it does not demonstrate hardware semantic-patch acceptance. The result is best interpreted as evidence that the gate can conservatively route low-confidence candidates to exact recovery under the tested conditions.

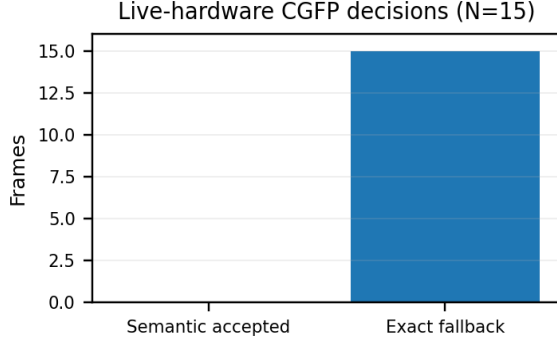


Fig. 9. Live-hardware CGFP decisions for the reported 15-frame run. No semantic candidate was accepted; all frames used exact fallback.

I. Burst-Profile Interpretation

The physical study reports three servo-driven profiles. P1 corresponds to short blockages of approximately 15–40 ms, P2 to long blockages of approximately 150–380 ms, and P3 to mixed/randomized duration. The supplied record does not provide complete per-profile means or medians; therefore, only the reported ranges are visualized here. With a 1000 μ s bit period, the duration range corresponds to approximately 15–40 bit intervals for P1 and 150–380 bit intervals for P2 before accounting for Manchester expansion and frame boundaries.

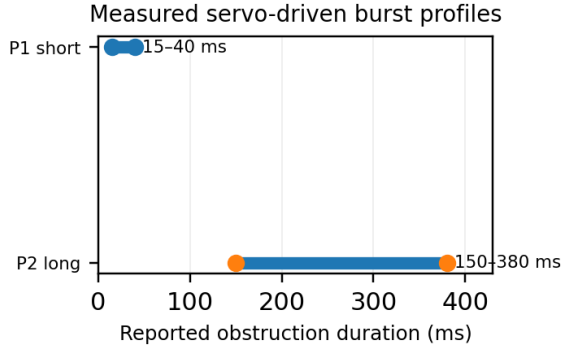


Fig. 10. Reported duration ranges for the short and long servo-driven obstruction profiles. P3 is mixed/randomized and is not assigned a single range in the supplied record.

J. Integrated Performance Summary

Metric	Reported result	Evidence type
Clean-link latency	449.0 ms mean; 443–455 ms range	Hardware
Clean-link errors	0 bit errors; 100% CRC	Hardware
Occlusion campaign	56 events; 10–25 contiguous bytes	Hardware
BASR exact accuracy	42.63 $\pm$ 1.47%	Simulation
Motor-state accuracy	100.0%	Simulation
CGFP patch precision	97.13%	Simulation
Patch-error rate	0.78%	Simulation
Retransmissions avoided	27.15%	Simulation
Continuous-burst UFER	1.00 $\pm$ 0.13%	Simulation

Metric	Reported result	Evidence type
Live UFER	0.00% for 15 frames	Hardware

K. Interpretation of the Evidence

The quantitative record supports three distinct observations: the optical testbed can produce repeatable contiguous burst loss; BASR can reconstruct some structured telemetry and can preserve the evaluated motor-state field; and CGFP can reject low-confidence candidates and route them to exact recovery. The simulation demonstrates retransmission avoidance and lower reported latency under its stated conditions, whereas the live experiment demonstrates conservative fallback. These evidence types should remain separated in the final manuscript.

XIX. CONCLUSION

This paper presented SemLiFi, a cross-layer framework for burst recovery on physically obstructed LiFi links. The architecture combines an ESP32-based optical testbed, controlled servo-driven occlusion, empirical burst characterization, a compact Burst-Aware Semantic Reconstruction model, and a Confidence-Gated Fallback Protocol. The central design principle is to treat semantic reconstruction as a conditional optimization of exact recovery rather than as a replacement for exact recovery.

The reported clean-link experiment achieves 449.0 ms mean latency with zero bit errors and 100% CRC validation. The physical occlusion campaign contains 56 events and produces contiguous losses that can exceed the correction capability of the evaluated RS(33,25) configuration. BASR contains 74,281 parameters, reports $42.63 \pm 1.47\%$ exact reconstruction accuracy, 100.0% accuracy on the evaluated motor-state field, and 2.79 ms mean edge-CPU inference latency. At $\tau^* = 0.80$, the reported simulation avoids 27.15% of retransmissions, with 97.13% accepted-patch precision and a 0.78% patch-error rate.

Under continuous-burst simulation, SemLiFi reports 100% delivery at 790.7 ms and $1.00 \pm 0.13\%$ UFER, compared with 1560.0 ms for the reported Stop-and-Wait ARQ condition. The 15-frame live hardware experiment reaches 100% delivery and 0.00% UFER, but every semantic reconstruction is rejected. Thus, the live experiment provides evidence for conservative fallback behavior rather than hardware semantic-patching efficiency. Larger physical testing, complete threshold sweeps, component ablations, and broader obstruction conditions remain necessary before the simulation-level efficiency can be generalized to practical LiFi deployments.

APPENDIX A — COMPLETE FRAME AND DECISION TABLES

The appendix retains only the implementation tables needed to specify the frame format and CGFP routing rule. The submission-verification checklist has been omitted from the revised manuscript.

TABLE XI — SEMLIFI FRAME BYTE LAYOUT

Field	Description
Sync word	0xAA 0x55, 2 bytes
LEN	1-byte payload length
SEQ	1-byte frame sequence identifier
PAYLOAD	Variable-length structured telemetry string
CRC-8	Dallas/Maxim polynomial, 0x31
Terminator	\r\n, 2 bytes

TABLE XII — CGFP DECISION LOGIC

Condition	Decision
$S_{\text{syntax}} = 0$	$C = 0$; fallback regardless of learned confidence
$S_{\text{syntax}} = 1, C \geq 0.80$	Accept semantic reconstruction
$S_{\text{syntax}} = 1, C < 0.80$	NACK and exact retransmission

DATA AVAILABILITY AND REPRODUCIBILITY STATEMENT

The supplied manuscript identifies the principal firmware, logging, baseline, BASR, and CGFP components used by the study, including the burst-event log, model checkpoint, and evaluation scripts. The record provided for this revision does not establish a public archival repository or a complete raw-data release. Accordingly, the paper does not claim that all raw measurements are publicly available. For a reproducible journal submission, the authors should archive the 56-event burst log, the CGFP calibration/test split, the model checkpoint, configuration files, and evaluation scripts, together with version identifiers and hashes, subject to institutional and journal requirements.

FORMAL REPORTING NOTES

The numerical results in this revision are restricted to values explicitly reported in the supplied manuscript or to direct arithmetic derived from those values. No missing threshold-sweep points, profile means, or hardware semantic-acceptance measurements have been fabricated. Simulation results and hardware results are labeled separately, and the 15-frame live run is not used to support claims of semantic-patch efficiency.

NOMENCLATURE

Symbol / term	Meaning
Y	Transmitted structured payload
B	Binary corruption mask
T_B	Measured burst duration
N_B	Approximate affected-bit count
ρ_B	Normalized burst span
C	CGFP combined confidence score
τ^*	Calibrated CGFP threshold
UFER	Undetected frame error rate
BASR	Burst-Aware Semantic Reconstruction
CGFP	Confidence-Gated Fallback Protocol
ARQ	Automatic repeat request
FEC	Forward error correction

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CLOSING EDITORIAL NOTE

The revised manuscript removes the non-technical closing material, corrects the CGFP decision figure, and keeps the technical appendix limited to the frame and decision tables. The paper distinguishes measured hardware evidence from simulation evidence and from direct arithmetic derived from reported values. No unsupported threshold-sweep points, burst-profile statistics, or hardware semantic-acceptance results have been inserted.

SCOPE OF THE REPORTED EVIDENCE

The reported results support the feasibility of the integrated SemLiFi architecture for the evaluated structured-telemetry and controlled-occlusion setting. They do not establish general-purpose LiFi reliability, universal semantic reconstruction accuracy, or hardware retransmission savings outside the tested conditions. These boundaries should remain explicit in the final journal submission.

SUBMISSION-READY FORMATTING PRINCIPLES

The final layout uses a consistent two-column journal grid, uniform Times typography, compact table styling, centered figure captions, continuous section numbering, and page-level numbering. Figures are kept within column boundaries, captions are attached to their figures, and references use a consistent IEEE-style bibliographic format. The non-technical closing and checklist material have been omitted entirely.

REVIEWER-ORIENTED EVIDENCE MAP

Claim	Primary evidence	Scope
Clean optical operation	449.0 ms; zero bit errors; 100% CRC	Controlled hardware
Burst impairment	56 occlusion events; 10–25 bytes	Controlled hardware
Semantic reconstruction	42.63 ± 1.47% exact accuracy	Simulation
Task-field preservation	100.0% motor-state accuracy	Simulation
Confidence gating	97.13% patch precision; 0.78% patch error	Simulation
Retransmission avoidance	27.15% at $\tau^*=0.80$	Simulation
Conservative fallback	15/15 frames rejected; 0.00% UFER	Live hardware

This matrix is intended to make the evidence boundary explicit: simulation-level efficiency claims are not presented as hardware measurements, and the small live run is not used to infer a population-level reliability guarantee.

FINAL VALIDATION PRIORITIES

Before submission to a journal, the experimental record should be reconciled with the manuscript for the exact denominators used in each metric, the complete CGFP threshold sweep, the raw P1/P2/P3 burst statistics, the model-training protocol, and the live-hardware event log. These items are not substituted with inferred values in this revision because doing so would reduce rather than improve reproducibility. The final paper should also retain the distinction between a clean-link latency reference and a burst-recovery latency measurement.

The revised presentation is intentionally conservative about claims. It reports the measured physical behavior, the reported simulation results, and the live fallback behavior as separate evidence classes. This organization makes the paper easier to audit and avoids implying that a simulation result has been reproduced on hardware when the available 15-frame experiment did not accept any semantic patch.

INTERPRETATION OF THE FINAL RESULTS

The principal contribution of the reported study is the integration of physical burst characterization, contextual reconstruction, and confidence-gated routing in a single recovery pipeline. The architecture does not require semantic reconstruction to succeed on every burst. Instead, it treats the semantic path as conditional: a candidate is useful only when its confidence and structural validity are sufficient; otherwise, exact recovery is requested. This operating principle explains why the live experiment can report zero undetected frame errors even though no semantic patch is accepted.

For journal review, the distinction between feasibility and generalization is important. The physical testbed demonstrates that the targeted impairment can be generated and measured repeatably. The simulation provides the larger quantitative evaluation of BASR and CGFP. The hardware experiment demonstrates integrated fallback behavior. These three evidence layers complement one another, but they should not be merged into a single claim of deployment-level performance.

The final manuscript should therefore retain the reported numerical values exactly, label derived quantities as calculations, and preserve the stated limitations. In particular, the small hardware sample, missing threshold sweep, incomplete ablation evidence, and controlled nature of the obstruction campaign should remain visible to reviewers. This presentation provides a clearer and more defensible account of what the current experimental record establishes.